

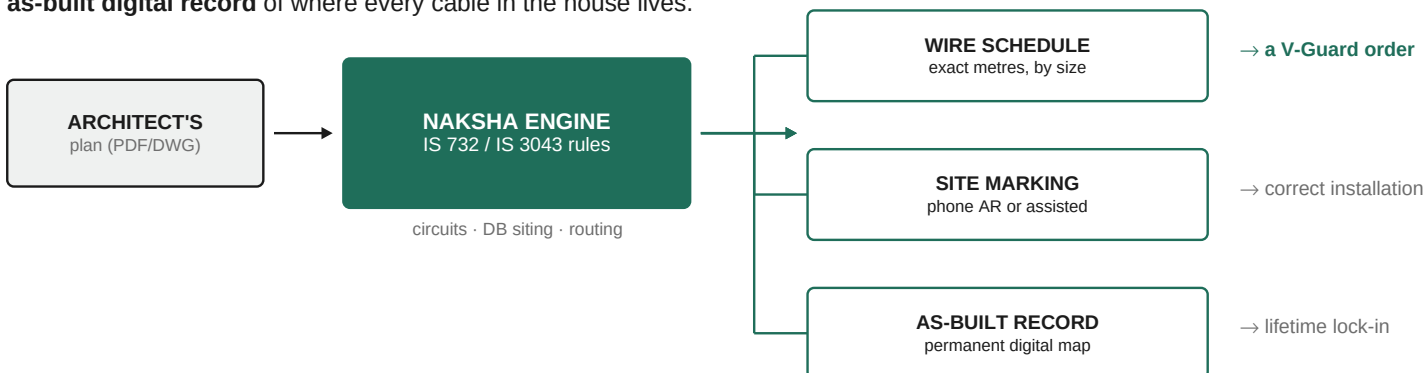
NAKSHA

V-Guard designs the wiring, then sells the wire

Track 4 · Reimagining V-Guard for an AI Powered Era · V-Guard Big Idea Tech Design Contest 2026

Wire is the most commoditised thing V-Guard sells. It competes on price, brand recall and whatever the electrician happens to have in his van. **NAKSHA changes what is being sold: not a coil of wire, but a designed electrical installation that happens to specify V-Guard wire.**

An owner, contractor or electrician uploads the architect's floor plan and answers a few questions about appliances. NAKSHA returns a complete residential electrical design — circuit split, distribution board siting, conduit routing, point placement — generated against **IS 732** and **IS 3043**. It produces three outputs: an exact **wire schedule in metres by size**, on-site **marking guidance** so the conduits actually go where the design says, and a permanent **as-built digital record** of where every cable in the house lives.



One upload, three outputs. The middle one gets the job built correctly; the first one is a purchase order; the third one never leaves the house.

The problem this actually solves

Indian homes are wired from memory. The electrician works without a drawing, improvises conduit routes, and leaves behind no record. The consequences are measurable: **roughly 42% of building fires in India are attributed to electrical short circuits**, and fire officials note that the phrase conceals ageing wiring, overloaded circuits, substandard material and years of neglect accumulating behind plaster.

The standards already exist. BIS publishes **IS 732** for wiring installation and **IS 3043** for earthing, and the CEA (Measures relating to Safety and Electric Supply) Regulations, 2023 require installation work to be carried out by a licensed electrical contractor under competent supervision. In February 2025 Parliament was asked why periodic household electrical safety checks are still not mandated.

So the gap is not the code. It is that nothing translates the code into a drawing anyone on an Indian residential site will actually use. And because there is no drawing, there is no as-built record — which is why people drill into live cables and why renovation means guessing.

The channel insight

Every major Indian wire brand has already digitised its electrician relationship. RR Kabel has RR Connect, Finolex has Samruddhi, Havells has an electrician app, and Polycab's Experts platform reaches around **2.5 lakh electricians and retailers**, which the company describes in its FY26 annual report as a grassroots influencer channel.

Every one of them is a loyalty scheme. Scan a QR code, collect points, redeem a gift.

The distribution channel is built, the electricians are on smartphones, and no brand has put a tool in that channel that does any actual work. An electrician does not want points. He wants the load calculation done, the wire sizes decided, the material list totalled, and a defensible answer when the owner asks why it costs what it costs.

WHAT EVERY COMPETITOR SHIPS

LOYALTY POINTS
scan QR · earn · redeem

RR Connect · Finolex Samruddhi · Polycab Experts · Havells

Polycab reaches ~2.5 lakh electricians this way.

WHAT NAKSHA SHIPS

A TOOL THAT DOES THE WORK
designs the job · sizes the wire

The channel is already digitised. Nobody has put anything useful in it.

The asset V-Guard needs already exists in the industry. Only its content is wrong.

Technology

- **Plan understanding.** Vectorise the uploaded plan; segment rooms, walls, doors and openings. Room-type inference from dimensions, adjacency and labels, with human confirmation rather than blind trust.
- **Rule engine.** IS 732 and IS 3043 encoded as explicit, auditable constraints — separate lighting and power circuits, dedicated ways for heavy loads, points-per-circuit limits, earthing requirements, diversity factors for load estimation. Deterministic and inspectable, not a language model guessing at safety rules.
- **Layout optimisation.** This is the core computational problem: place the distribution board and route conduits to minimise total wire length and voltage drop, subject to the rule set and to buildable chase geometry. A graph problem over the room adjacency network, solved with shortest-path routing plus a metaheuristic over board placement.
- **Bill of quantities.** Wire length per size with waste allowance, conduit, boxes, switchgear, mapped directly onto V-Guard SKUs.
- **Site delivery.** Phone-based AR overlays the design onto the actual wall using the plan geometry and device pose. Cheap, works on any modern handset, and requires no equipment on site.
- **As-built capture.** The electrician confirms or corrects each run as it is laid, before plastering. The corrected model becomes the homeowner's permanent record and the training signal that improves the generator.

Novelty, stated honestly

The components are not new and the proposal is stronger for saying so. Automated electrical takeoff is a mature commercial category — tools such as drawer.ai already accept PDF drawings and export quantities, device properties and **wire lengths with marked-up routing**, and Kreo, Trimble Accubid and PlanSwift serve the same market. Academic work presented at ISARC 2026 automates outlet-to-circuit and circuit-to-panelboard assignment in a CAD-BIM workflow. Robotic site layout is also solved: Dusty Robotics' FieldPrinter has printed coordinated models on more than **300 million square feet** of slab.

All of that serves **Western commercial contractors bidding large projects**. None of it serves an electrician wiring a 3BHK in Kochi, none of it encodes Indian standards, and none of it is distributed by the company that sells the wire.

Our contribution is therefore positional, not algorithmic: **Indian code compliance as the rule set**, **residential scale** where no drawing exists today rather than commercial scale where one already does, **distribution through a channel V-Guard already owns**, and the **as-built record** as a durable asset none of the takeoff tools produce because their users leave the site and never return.

Business value

- **Every design is a purchase order.** The output is a wire schedule in metres by size. Specifying the material is a categorically stronger position than advertising to whoever walks into a shop — this is the Asian Paints move, which used Beautiful Homes services to escape commodity paint competition, and which Berger followed with Express Painting.
- **It captures the decision years earlier.** Wiring is specified during construction, before the owner has any brand opinion, and it is never re-shopped. Rewiring a finished house is unthinkable.
- **It creates a portfolio funnel.** The same plan positions the water heater, chimney, RO and gas point — all V-Guard categories, all decided at construction stage, all currently decided by whoever is standing there.

- **It is a margin story, not a volume story.** V-Guard's FY25-26 revenue grew 7.0% to Rs 5,966 crore while PAT fell 1.7% to Rs 308 crore. Commodity wire cannot fix that; specified systems can.
- **The as-built record is the lock-in.** A household whose wiring map lives in V-Guard's app has a reason to return for every future addition, fault and renovation.

On scale — and the marking problem

A national service business is genuinely hard: staff, training, equipment, quality control. We do not propose one. The design engine scales at software cost; only the physical marking needs people, so we separate them deliberately.

Tier 1 · App	Free, all-India phone AR marking, zero hardware, infinitely scalable
Tier 2 · Assisted	Metro cities trained V-Guard marking partner, paid service
Tier 3 · Flagship	Showcase only robotic / laser layout — built for marketing, not margin

Tier 1 carries the strategy — free, national, zero marginal cost, and it is where the wire schedules come from. Tier 3 exists to be photographed. Treating the flagship as a marketing instrument rather than a P&L line is the honest position, and it is exactly how paint companies launched their service brands.

Risks and honest limitations

- **Plan quality is the hard input problem.** Indian residential plans are often scanned, hand-marked, incomplete or dimensionally unreliable. The engine must degrade gracefully to guided manual tracing, and the first release should assume human confirmation of every inferred room.
- **Liability.** Software that produces an electrical design in a jurisdiction where a licensed contractor is legally responsible must position itself as a decision aid the contractor signs off, not as the designer of record. This needs legal structuring before launch.
- **Electrician adoption is the real commercial risk,** not the technology. Experienced electricians may see the tool as a challenge to their judgment. It has to visibly save them time on load calculation and estimation on day one, or it will be ignored.
- **AR marking accuracy on a construction site** is unproven at the tolerances conduit chasing needs. Tier 1 should therefore promise guidance and dimensioned reference marks, not millimetre placement.
- **Track fit.** This is a Track 4 entry on the basis that the brief invites solutions “aligned with its core categories,” and wires and switchgear are core. We note plainly that the intelligence sits in the design tool rather than inside a powered product — though its output, the as-built model, becomes a permanent fixture of the home.

Phase 3 plan

- **Working layout generator** on real Indian floor plans: rooms in, circuits and routed conduits out, with the IS 732 rule set explicit and each decision traceable to a clause.
- **Optimisation study:** total wire length and voltage drop versus distribution-board placement across several house typologies — quantifying the material saving a good layout produces over an improvised one.
- **Interactive browser demo** generating and visualising a design from an uploaded plan, plus a costed bill of quantities.
- **AR feasibility test:** phone-based overlay on a real wall, with measured registration error reported honestly rather than asserted.